

RAG-Based Customer Support Assistant Using LangGraph & HITL

High-Level Design (HLD)

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Batch: Feb 2026

Organization: Innomatics Research Labs

Date: 23/04/2026

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High-Level Design (HLD)

RAG-Based Customer Support Assistant (LangGraph + Groq)

1. System Overview

Problem Definition

Customer support systems often rely on static FAQs, which lack contextual understanding. This project builds a dynamic assistant that retrieves relevant information from a knowledge base and generates accurate responses.

Scope

- Accept user queries via UI
 - Process PDF knowledge base
 - Retrieve relevant information using embeddings
 - Generate contextual answers using LLM
 - Support Human-in-the-Loop (HITL) escalation
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2. Architecture Overview

Main Components

- **User Interface (Streamlit)**
Accepts user queries and displays responses
- **Document Ingestion Pipeline**
Loads and processes PDF documents
- **Chunking Module**
Splits document into meaningful sections
- **Embedding System**
Converts text into vector representations
- **Vector Database (ChromaDB)**
Stores embeddings for fast retrieval
- **Retriever**
Fetches relevant chunks based on query

- **LLM (Groq - LLaMA3)**
Generates contextual answers
 - **Workflow Engine (LangGraph)**
Controls execution flow
 - **Routing Layer**
Decides between answering or escalation
 - **HITL Module**
Handles escalation to human agent
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3. Data Flow

1. User uploads PDF
 2. PDF is split into sections
 3. Sections are embedded using embedding model
 4. Embeddings stored in ChromaDB
 5. User submits query
 6. Retriever fetches relevant chunks
 7. LLM generates response using context
 8. Routing decides:
 - Answer → return to user
 - Escalate → HITL
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4. Component Description

- **Document Loader:** Loads PDF content
 - **Chunking Strategy:** Section-based splitting using numbered headings
 - **Embedding Model:** HuggingFace embeddings
 - **Vector Store:** ChromaDB
 - **Retriever:** Similarity-based retrieval
 - **LLM:** Groq (LLaMA3 model)
 - **Graph Workflow Engine:** LangGraph
 - **Routing Layer:** Confidence + response-based logic
 - **HITL Module:** Escalates unresolved queries
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5. Technology Choices

- **ChromaDB**
Lightweight and easy vector database
- **LangGraph**
Enables graph-based workflow and decision routing

- **Groq API**
Fast and efficient LLM inference
 - **Streamlit**
Simple UI for interaction
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6. Scalability Considerations

- Handles large documents via chunking
- Retrieval optimized using top-k search
- Modular design allows scaling to APIs or web apps
- Can extend to multi-document support